ZIAD EMAD ALLAM

Software Developer

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Software Developer delivering data-heavy React/Next.js apps and secure Node/Express APIs. Shipped 10+ MERN projects. Built analytics dashboards (+15% engagement) and CI/CD pipelines (−70% deployment time). Created Python QA tooling for embedded systems, replacing LabVIEW (−$30k/yr.) and achieving 86% test automation. Deployed LLM features deflecting ~30% support tickets and improving response accuracy by 25%.

WORK EXPERIENCE

R&D Software Engineer (Full-time) | **Elaraby Group** ***06/2024 – Present***

- Built Python/AI QA tooling for embedded firmware—reusable test harnesses, automated data acquisition, and signal/sensor analytics (NumPy, SciPy, Pandas, FFT, filtering, anomaly detection).
- Replaced LabVIEW with Python pipelines saving $30,000/yr; raised test automation from 40% to 86%.
- Integrated with Dassault 3DEXPERIENCE (PLM) for traceable defect reporting and faster triage via interactive visuals.

Full-Stack Developer (Freelance) | **Upwork** ***11/2022 –11/2025***

- Engineered 10+ full-stack MERN projects for global clients, achieving a 4.8/5 satisfaction rating
- Developed analytics dashboards leading to a 15% increase in user engagement for a key client's app
- Implemented CI/CD pipelines reducing manual deployment time by 70%.

Software Engineer (Part-time) | **OpenQQuantify** ***02/2025 – 11/2025***

- Deployed LLM-powered features automating 30% of customer support queries, improving response time.
- Engineered and optimized system prompts to enhance AI response accuracy by 25%.

PROJECTS

ZEDX AI Copilot – Real-Time AI Meeting & Accessibility Assistant **GitHub** ↗ **Live** ↗

Tech: Next.js, React, TypeScript, Electron, Tailwind, Supabase, Groq API, WebRTC, Tesseract.js

- Engineered a scalable, cross-platform desktop application using Next.js and Electron, delivering a highly responsive AI meeting assistant that served thousands of users.
- Developed real-time, low-latency audio processing utilizing WebRTC, Media Stream APIs, and Voice Activity Detection (VAD), seamlessly integrated with Groq Whisper to achieve 99%-accurate speech-to-text conversion.
- Integrated LLaMA 3, GPT, Qwen models and client-side OCR (Tesseract.js) to provide context-aware insights from user-uploaded documents (Agendas/Resumes) and enable instant text extraction.
- Architected a secure and scalable backend with Supabase (PostgreSQL), utilizing strict TypeScript data models and Row Level Security (RLS) policies to significantly reduce runtime errors and protect historical user data.
- Built an interactive and accessible React-based UI optimized with Tailwind CSS and Framer Motion, featuring clean overlay modes.

Enterprise QA Automation Framework (Microwave HIL System) **GitHub** ↗

Tech: Python, PyQt, nidaqmx, State Machines, openpyxl, Asynchronous Programming

- Architected an industrial-grade automation framework using Python and asynchronous PyQt interfaces, processing high-frequency real-time DAQ telemetry without compromising UI responsiveness.
- Engineered deterministic State Machines and complex analytical algorithms to autonomously validate hardware behavior and enforce critical safety protocols (e.g., preventing catastrophic signal overlaps).
- Developed automated data pipelines that dynamically translate raw multi-channel telemetry into comprehensive, highly formatted Excel diagnostic matrices using openpyxl.

Enterprise Automation & AI Diagnostics Framework **GitHub** ↗

Tech: Python, ML, NumPy, SciPy, Pandas, PyQt6, nidaqmx, DSP, Docker, CI/CD

- Transitioned legacy LabVIEW to Python framework: $30k annual cost reduction, test coverage 40%→86%; engineered bidirectional DAQ pipelines for real-time IEC compliance diagnostics.
- Integrated AI predictive models and FFT-based DSP algorithms in async PyQt6 interface for zero-latency defect triage.

TECHNICAL SKILLS

Frontend: HTML5, CSS3, Tailwind CSS, JavaScript, TypeScript, React, Next.js, Vue, Angular, Framer Motion
Backend: Node.js, Express, Python (FastAPI, Flask), PHP, REST, JWT/RBAC, WebSockets/Socket.IO
Data & AI: NumPy, SciPy, Pandas, Matplotlib, n8n, LLMs (AI Agents), Prompt Engineering, TensorFlow
DevOps: Git, GitHub, CI/CD, AWS, Supabase, Docker, Selenium, Pytest
Databases: MongoDB, PostgreSQL, MySQL, SQLite
Analysis: Python (Tkinter, PyQt, PySide), DSP, Signal Processing

EDUCATION

Bachelor of Computer and Information Systems | **Tanta University, Egypt** ***10/2023 – 06/2027***

Coursework: Data Structures, Algorithms, Software Development

Diploma in Data Analysis | **The American University in Cairo** ***07/2024 – 07/2025***

Coursework: Advanced Python | GPA: 3.8

CERTIFICATIONS & LANGUAGES

Certifications: **AUC** Data Analysis Diploma · **Oracle** Cloud Gen AI Professional · Harvard CS50 · **micro1** Certified SWE · **Cisco** JS · **Microsoft** (Azure OpenAI, API Mgt, Cosmos DB, Blob Storage, Auth MSAL) · **DeepLearning**.AI Agile Tester · **DataCamp** Python Certified.

Languages: Arabic (C2 – Native) · English (C2 – Fluent)